

# ZX Spectrum 48K → PC Keyboard Map

All 40 Spectrum keys map one-to-one onto a standard 104/105-key PC keyboard. Legends below are extracted from the 48K ROM's own key tables, so they match what the machine actually produces — not what is silk-screened on any particular case.

## THE SPECTRUM KEYBOARD



Above the key — BASIC keyword (K mode), or CAPS SHIFT function on the digits    On the key, red — SYM SHIFT    Below, green — EXTENDED mode (on digits: a colour, as PAPER; as INK with CAPS SHIFT)

Below, red — EXTENDED + SYM SHIFT

## THE SAME KEYS ON A STANDARD PC KEYBOARD



Mapped — wired in `keyb_lookup.h` today    Left Shift = CAPS SHIFT, Right Shift = SYM SHIFT    Combination key — exists on the Spectrum only as SHIFT + key (see overleaf)    No equivalent

# Matrix Wiring & Shifted Combinations

The keyboard is an 8×5 matrix read through the ULA. Any port with bit 0 clear selects it; the high byte of the port chooses which rows to scan. Rows are active *low*, and several may be selected at once, in which case their results are ANDed — that is how the ROM detects a shifted key. A pressed key reads back as **0**.

## MATRIX MAP

| ROW | ADDRESS LINE | PORT   | BIT 0                           | BIT 1                           | BIT 2    | BIT 3    | BIT 4    |
|-----|--------------|--------|---------------------------------|---------------------------------|----------|----------|----------|
| 0   | A8           | 0xFEFE | <b>CAPS SHIFT</b><br>Left Shift | <b>Z</b>                        | <b>X</b> | <b>C</b> | <b>V</b> |
| 1   | A9           | 0xFDFE | <b>A</b>                        | <b>S</b>                        | <b>D</b> | <b>F</b> | <b>G</b> |
| 2   | A10          | 0xFBFE | <b>Q</b>                        | <b>W</b>                        | <b>E</b> | <b>R</b> | <b>T</b> |
| 3   | A11          | 0xF7FE | <b>1</b>                        | <b>2</b>                        | <b>3</b> | <b>4</b> | <b>5</b> |
| 4   | A12          | 0xEFFE | <b>0</b>                        | <b>9</b>                        | <b>8</b> | <b>7</b> | <b>6</b> |
| 5   | A13          | 0xDFFE | <b>P</b>                        | <b>O</b>                        | <b>I</b> | <b>U</b> | <b>Y</b> |
| 6   | A14          | 0xBFFE | <b>ENTER</b><br>Enter           | <b>L</b>                        | <b>K</b> | <b>J</b> | <b>H</b> |
| 7   | A15          | 0x7FFE | <b>SPACE</b><br>Space           | <b>SYM SHIFT</b><br>Right Shift | <b>M</b> | <b>N</b> | <b>B</b> |

**Rows 3 and 4 run in opposite directions.** Row 3 counts 1 2 3 4 5 outward from the left edge of the keyboard and row 4 counts 0 9 8 7 6 outward from the right, mirroring the physical wiring. This is a classic source of transposed-key bugs.

## SHIFTED COMBINATIONS A PC USER EXPECTS

The Spectrum has no dedicated Backspace, cursor, comma or period keys — they are all SHIFT combinations. To support the PC keys below, one scancode must pull *two* matrix bits low.

| PC KEY           | CAPS SHIFT + | MATRIX BITS   | PRODUCES     | CHAR | SYM SHIFT + | MATRIX BITS   | CHAR | SYM SHIFT + | MATRIX BITS   |
|------------------|--------------|---------------|--------------|------|-------------|---------------|------|-------------|---------------|
| Backspace        | 0            | (0,0) + (4,0) | DELETE       | ,    | N           | (7,1) + (7,3) | /    | V           | (7,1) + (0,4) |
| ← Left           | 5            | (0,0) + (3,4) | cursor left  | .    | M           | (7,1) + (7,2) | *    | B           | (7,1) + (7,4) |
| ↓ Down           | 6            | (0,0) + (4,4) | cursor down  | ;    | O           | (7,1) + (5,1) | ?    | C           | (7,1) + (0,3) |
| ↑ Up             | 7            | (0,0) + (4,3) | cursor up    | '    | 7           | (7,1) + (4,3) | :    | Z           | (7,1) + (0,1) |
| → Right          | 8            | (0,0) + (4,2) | cursor right | "    | P           | (7,1) + (5,0) | @    | 2           | (7,1) + (3,1) |
| Caps Lock        | 2            | (0,0) + (3,1) | CAPS LOCK    | -    | J           | (7,1) + (6,3) | £    | X           | (7,1) + (0,2) |
| Ctrl (suggested) | SPACE        | (0,0) + (7,0) | BREAK        | =    | L           | (7,1) + (6,1) | <    | R           | (7,1) + (2,3) |
|                  |              |               |              | +    | K           | (7,1) + (6,2) | >    | T           | (7,1) + (2,4) |

**On the digit keys, extended mode sets a colour, not a keyword.** EXTENDED then a *letter* gives that letter's keyword, but EXTENDED then a *digit* inserts a colour change into the line: digits 0-7 set the **PAPER** of the text that follows, or its **INK** with CAPS SHIFT; 8 and 9 give BRIGHT 0/1, or FLASH 0/1 with CAPS SHIFT. Inside a PRINT string that colours individual words. The digit keywords ( DEF FN , LINE , FORMAT ...) need EXTENDED + SYM SHIFT + digit, and the colour keywords themselves are on letters: INK X, PAPER C, FLASH V, BRIGHT B, INVERSE M, OVER N (each EXTENDED + SYM SHIFT), TAB P (EXTENDED) and AT I (SYM SHIFT alone).

**This is a positional map, not a character map.** Shift + 2 prints @ on a PC, but CAPS SHIFT + 2 on a Spectrum is CAPS LOCK — @ is SYM SHIFT + 2. Mapping physical positions keeps the emulator faithful; a "PC-friendly" typing mode is a separate translation layer.

Legends extracted from 48.rom: token table 0x0095; SYM SHIFT letters 0x026A; EXTENDED letters 0x022C; EXTENDED+SYM letters 0x0246; CAPS SHIFT digits 0x0260; EXTENDED + SYM digits 0x0284. K-mode keywords are token 0xE6 + (letter - 'A'), computed rather than tabulated. EXTENDED digits are decoded by SUB \$20 / SUB \$36 at 0x037B and then turned into PAPER/INK/BRIGHT/FLASH by the editor — verified by printing text after each combination and reading back its attribute. SYM SHIFT digit symbols and every extended-mode path verified by running the ROM and reading LAST-K ( 0x5C08 ). Matrix assignments mirror include/z80/keyboard\_lookup.h.